Risk-Based Pricing & Underwriting Framework for Student Renter's Insurance

CAS Case Competition

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Executive Summary

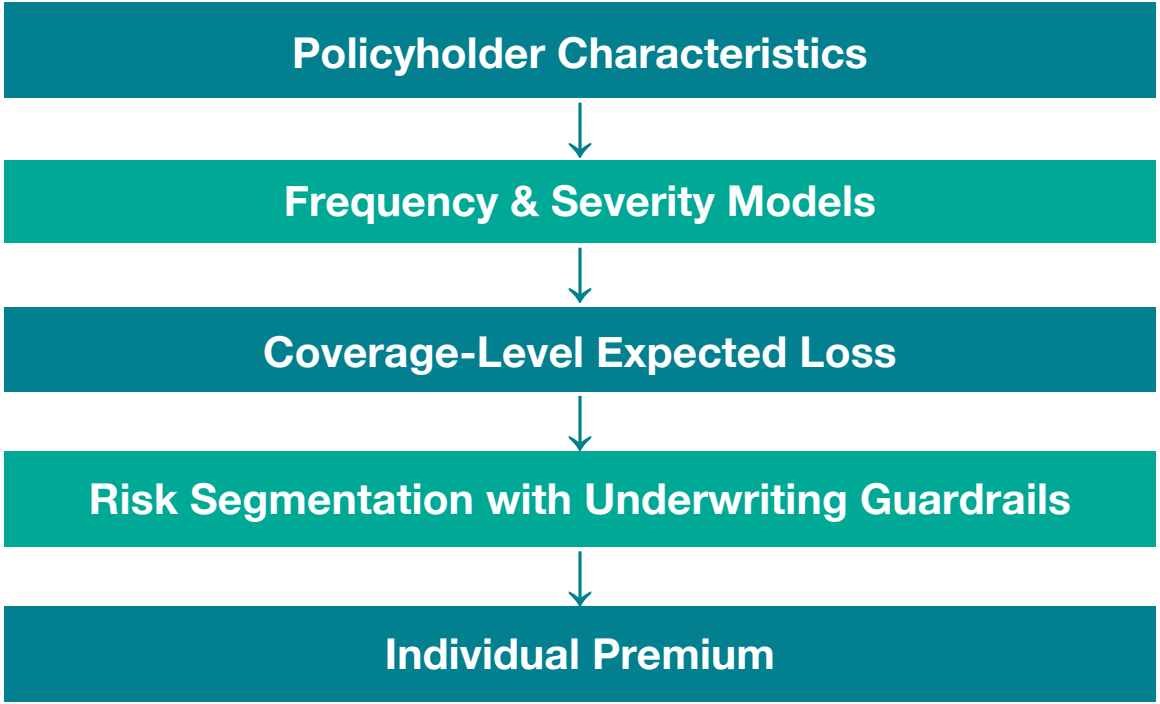
OBJECTIVE

Develop a data-driven pricing framework for student renters' insurance

SOLUTION

- Combine predictive modeling with risk segmentation and underwriting controls
- Price individual risk rather than portfolio averages
- Results in more **stable loss ratios** and **improved profitability**

Overall Pricing Framework



Outputs: Individual premium, risk tier, underwriting action

Data & Input

Data Structure

Policy-level dataset with unit exposure, claim counts, and claim amounts

Modeling Scope

Suitable for frequency and severity modeling

Preprocessing

Data cleaning (duplicates, harmonization, outliers, missing values), categorical encoding, mild response capping

Limitations & Enhancements

Additional information such as exposure duration and historical policy experience would improve accuracy

Frequency Modeling - Summary

OBJECTIVE & METHOD

- Objective: Estimate expected number of claims per insured
- Method: Poisson GLM (separate models by coverage)

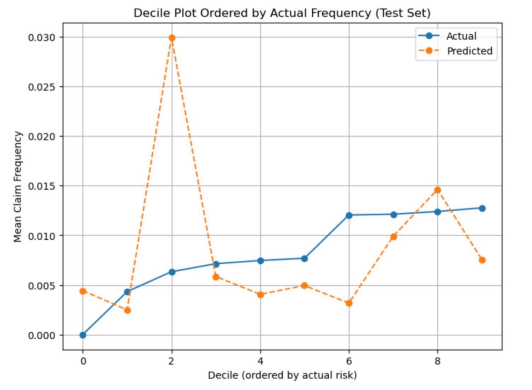
KEY ASSUMPTIONS

- Unit Exposure
- Claims independent across observations
- Poisson counts (checked for overdispersion)

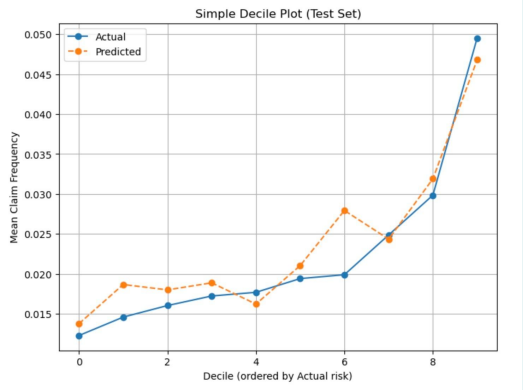
Model Development & Validation:

- Model Choice: AIC comparison; selected interactions; partial residual plot
- Variable Selection: Sensitive variables excluded for fairness
- Applicability: Valid under similar future risk profiles; re-estimate if covariate distributions shift
- Validation: Hold-out decile calibration

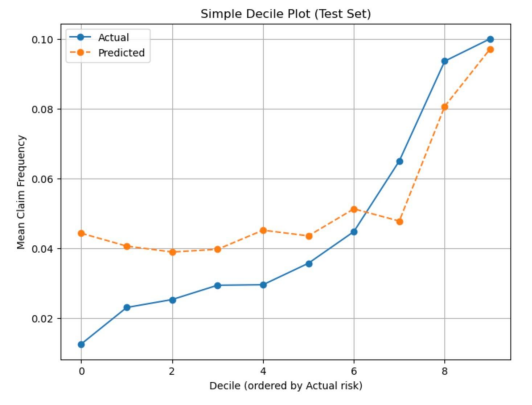
Frequency Modeling - Validation Charts



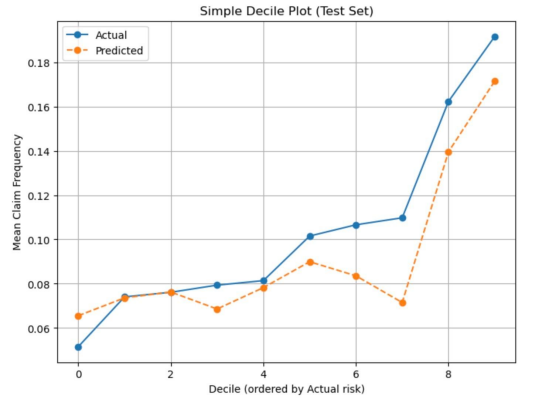
Liability



Guest Medical



Additional Living Expense

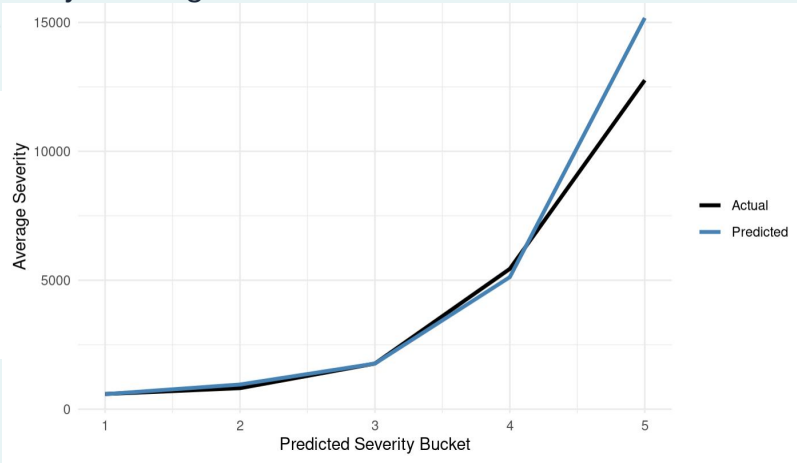


Personal Property Protection

Severity Model

OBJECTIVE & METHOD

- Objective: Estimate average claim size conditional on a claim by coverage
- Method: Gamma GLM with log link separated by coverage



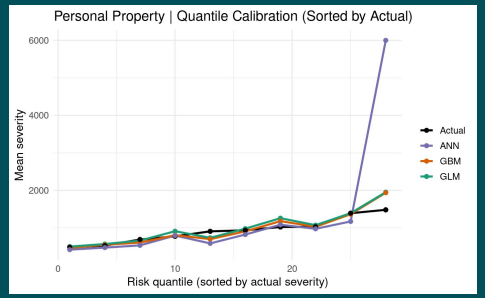
Rationale:

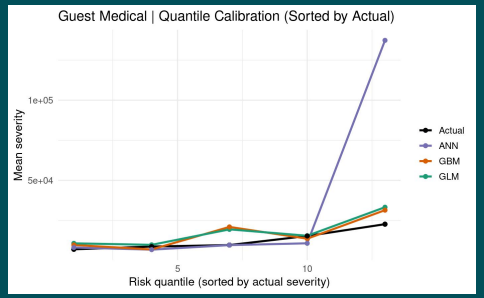
- Positive, right-skewed loss amounts
- Multiplicative effects align with insurance pricing practice

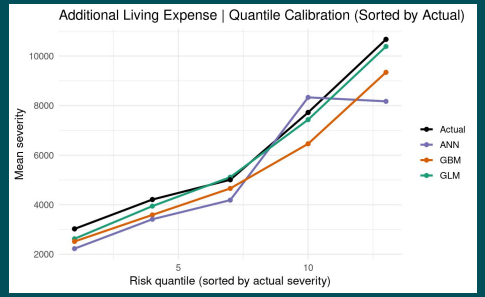
Results:

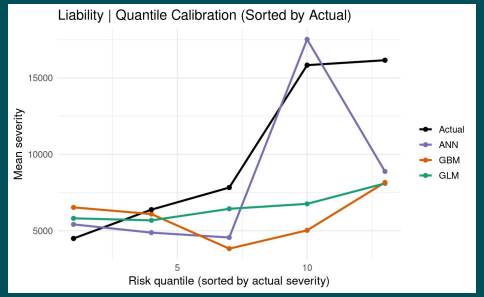
Quantile-based calibration comparing predicted and observed severity across the holdout set shows good alignment in higher-risk quantiles — the model preserves risk ordering rather than just fitting averages.

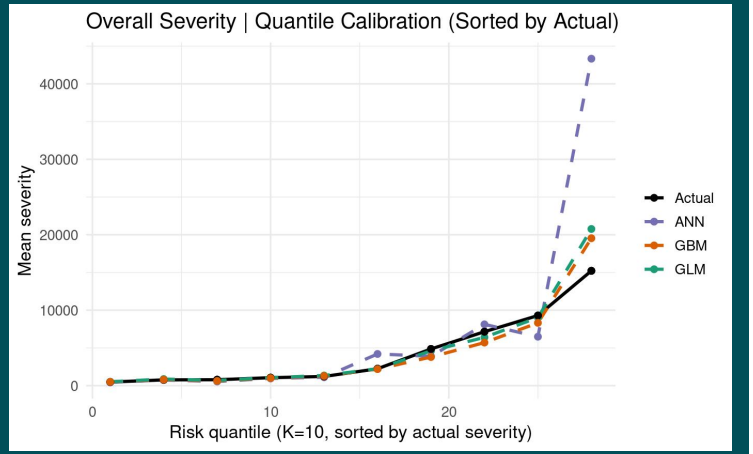
Severity Model - ML











KEY INSIGHTS

- GLM & GBM Show Near-Identical Performance
- ANN Underperforms in High-Risk Quantiles
- Overall: Stability & Interpretability Trump Marginal Gains

MODEL STRATEGY

- Prioritize GLM
- Use GBM as Validation Tool
- Exercise Caution with ANN/Black-Box Models

Expected Loss - Core Risk Measure

COVERAGE LEVEL

For coverage c in
C = {Personal Property, Liability, Guest Medical, Additional Living Expense}:

$$Expected\ Loss_c = Frequency_c \times Severity_c$$

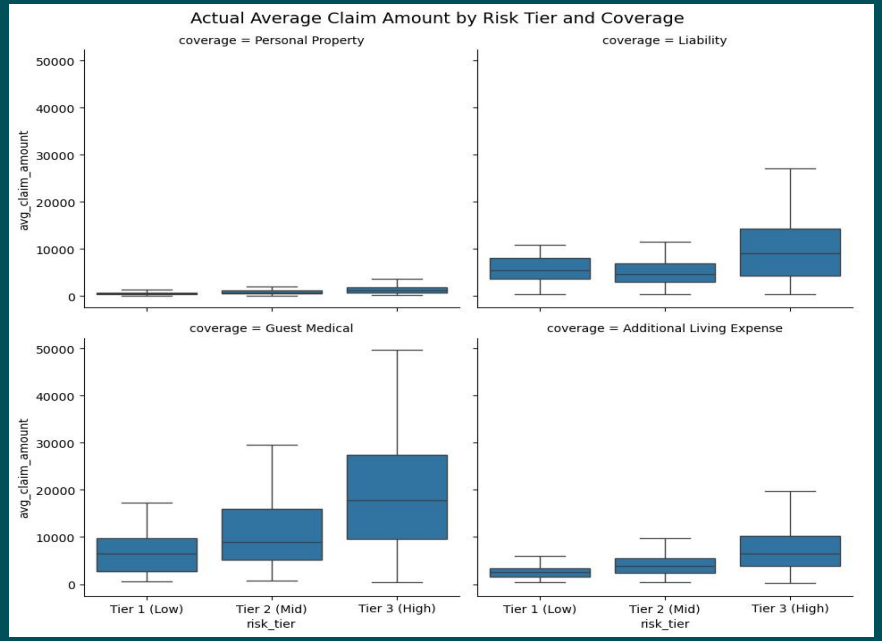
$$Total\ Expected\ Loss = \sum_{c}^{C} Expected\ Loss_c$$

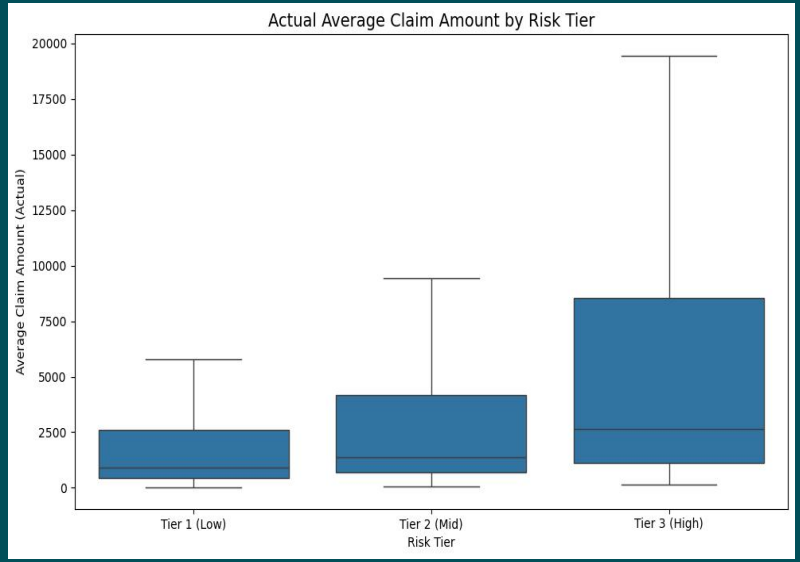
Risk Segmentation

- Policyholders ranked by predicted total expected loss
- Risk segmented into three tiers reflecting loss concentration
- Enables differentiated pricing and underwriting decisions

Risk Tier	Portfolio Share	Average EL	Relativity
Preferred	40%	$307	0.52x
Standard	30%	$594	1.00x
Non-Standard	30%	$1868	3.14x

Risk Segmentation - Effectiveness





Segmentation effectiveness is evaluated using actual historical claim experience.

Credits & Debits within Tier

PURPOSE

Refine pricing without excessive complexity

FRAMEWORK

- Applied only to stable, interpretable characteristics
- Adjustments do not change tier

Governance Principles:

- Applied only within tier
- Based on directional model evidence
- Subject to hard caps to avoid over-segmentation (+-5%)

Characteristic	Condition	Adjustment
Fire Protection	Sprinklered	-3% credit
Housing	On-campus	-2% credit
Loss History	Prior claim	+3% debit

Premium

$$Premium = \frac{EL}{1 - Expense\ Ratio\ -\ Profit\ Margin}$$

Pricing Assumptions:

- Expense Ratio: ~35% (loss adjustment expenses + underwriting expenses)
- Profit Margin: ~5% (risk and capital return)

Risk Tier	Average EL	Premium
Preferred	$307	$511.67
Standard	$594	$990
Non-Standard	$1868	$3113.33

Transparent and implementable pricing.

New vs Renewal Strategy

NEW BUSINESS DISCOUNTS

- Preferred: 8-10%

Benchmarked: State Farm, Nationwide

- Standard: 5-7%

Benchmarked: Progressive, GEICO

- Non-Standard: 3-5%

RENEWAL LOYALTY DISCOUNTS

- Year 1: 5%
- Year 3: 8-10%
- Year 5+: Up to 15%

Benchmarked: Nationwide's program

Strategy: Balance acquisition investment with escalating loyalty rewards to maximize customer lifetime value

Pricing Strategy Over Time

Monte-Carlo used to stress-test our pricing strategy under extreme but plausible scenarios

This strategy avoids overpricing low-risk years while protecting the company against catastrophic outcomes.

Key Findings:

01 **Obtain Tail Risk Protection Without Overpricing**

02 **Monitor Structural Shifts, Not Volatility, Trigger Reassessment**

03 **Achieve Risk Tier Alignment with Premiums**

	coverage	risk_tier	n_policies	freq_kind	sev_kind	phi_gamma	mean_total_loss	median_total_loss	p75_total_loss	p90_total_loss	p95_total_loss	p99_total_loss	VaR95	TVaR95	VaR99	TVaR99	pure_premium_per_policy	gross_premium_per_policy
0	Additional Living Expense	1	910	poisson	gamma	0.321574	240799.533835	238906.360016	269556.925102	299040.012177	317722.069284	353797.648444	317722.069284	340095.250971	353797.648444	373416.177900	264.614872	304.307103
1	Additional Living Expense	2	439	poisson	gamma	0.321574	125118.315807	122779.508671	146607.243146	171010.633159	185801.304455	215717.382162	185801.304455	204180.172295	215717.382162	231438.688869	285.007553	327.758686
2	Additional Living Expense	3	665	poisson	gamma	0.321574	192065.895414	189765.382646	219304.796506	248635.891418	266592.112506	302560.730860	266592.112506	289268.727511	302560.730860	323484.033258	288.820895	332.144030
3	Guest Medical	1	890	poisson	gamma	0.773477	984.062814	0.000000	0.000000	0.000000	0.000000	36637.071992	0.000000	984.062814	36637.071992	74297.123151	1.105689	1.271542
4	Guest Medical	2	495	poisson	gamma	0.773477	118.115734	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	118.115734	0.000000	118.115734	0.238618	0.274410
5	Guest Medical	3	628	poisson	gamma	0.773477	96.237144	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	96.237144	0.000000	96.237144	0.153244	0.176230
6	Liability	1	886	poisson	gamma	0.633353	150.549056	0.000000	0.000000	0.000000	0.000000	3794.024271	0.000000	150.549056	3794.024271	14527.590877	0.169920	0.195408
7	Liability	2	459	poisson	gamma	0.633353	7.917147	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	7.917147	0.000000	7.917147	0.017249	0.019836
8	Liability	3	656	poisson	gamma	0.633353	11.128210	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	11.128210	0.000000	11.128210	0.016964	0.019508
9	Personal Property	1	913	poisson	gamma	0.349659	79844.983711	79444.375363	86991.563135	94117.010746	98580.890176	106856.762891	98580.890176	103777.279290	106856.762891	111297.879736	87.453432	100.571447
10	Personal Property	2	422	poisson	gamma	0.349659	38415.252585	37932.042901	43570.080310	48835.877397	52127.868009	58707.356569	52127.868009	56106.580319	58707.356569	61921.302303	91.031404	104.686115
11	Personal Property	3	638	poisson	gamma	0.349659	62736.908708	62216.770271	70022.087345	77321.039111	81794.815572	90360.087364	81794.815572	87120.246331	90360.087364	95405.088991	98.333713	113.083770

Pricing Strategy Over Time: Re-rating Strategy Framework

Core Annual Cycle:

1 **Model Re-estimation & Stability Check**

Refit frequency and severity models using latest exposure and loss data to ensure alignment with current risk profiles

2 **Monte Carlo Simulation Refresh**

Update economic scenarios to monitor actionable risk signals and emerging tail risk trends

3 **Tail Risk Benchmarking (P&C & Specialty Lines)**

TVaR95/Mean Ratio	Interpretation & Action
1.2 – 1.5	Acceptable risk profile; no immediate action needed
≥ 2	Re-examine pricing structure and risk assumptions
≥ 3	**Clear tail-driven business; urgent pricing/model review required**

Expense Considerations

BEST PRACTICE

Expenses should ideally be allocated by underlying cost drivers, rather than modeled as a single percentage of premium.

Examples of Cost Drivers:

- Exposure: For underwriting & servicing
- Claim counts or losses: For claims adjusting

SIMPLIFIED APPROACH IN THIS PROJECT

A single overall expense loading was applied to preserve the integrity of relative risk ranking. Future enhancements would incorporate driver-based allocation.

Final Recommendation

- Implement tier-based pricing driven by expected loss
- Improve portfolio **stability** by controlling tail risk through underwriting guardrails
- Apply limited within-tier credits and debits to enhance fairness and **retention**
- Use simplified expense assumptions for rate level, refined over time
- Monitor loss ratio and volatility by tier and **recalibrate annually**

- **More predictable loss ratios**
- **Improved profitability**
- **Clear actions for underwriting and pricing decisions**

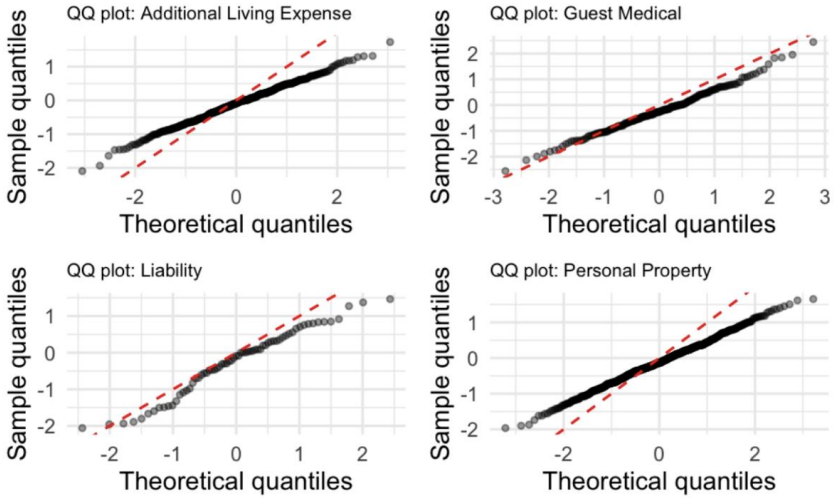
APPENDIX

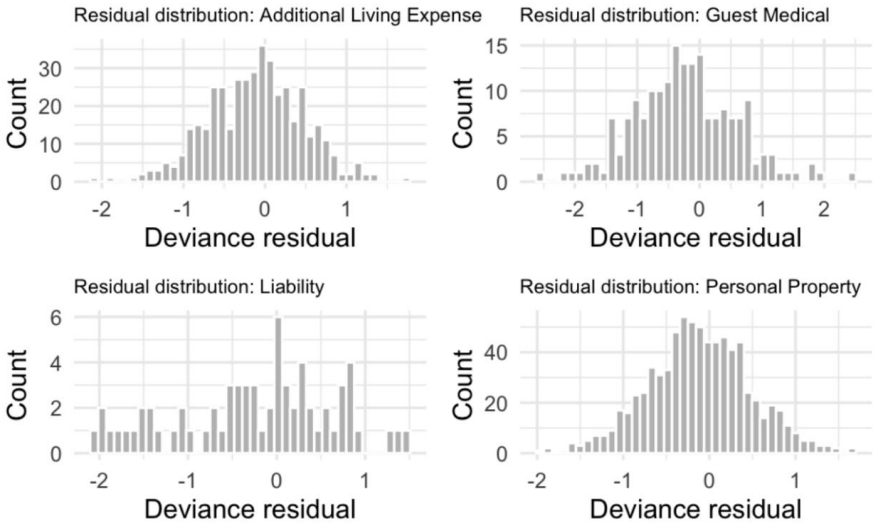
Additional Technical Details & Supporting Analysis

Appendix: Overdispersion Checks for Frequency

	coverage	n	mean	var
0	Additional Living Expense	10000	0.0528	0.053217
1	Guest Medical	10000	0.0233	0.023559
2	Liability	10000	0.0085	0.008429
3	Personal Property	10000	0.0973	0.098643

Appendix: Assumption Checks for Severity





Appendix: Detailed Statistical Modeling Techniques

Capping & Transformation	Capping applied to selected variables to mitigate extreme values and improve robustness. Partial residual plots used to assess functional form.
Interaction Terms	Greek affiliation and off-campus residence interaction included based on actuarial reasoning, reflecting social and exposure pattern differences.
Model Selection	AIC used to compare candidate specifications from alternative interactions and discretizations of continuous predictors.
Robustness Checks	Cook's distance and bootstrap resampling confirmed stable parameter estimates. VIFs assessed multicollinearity—all within thresholds.
Validation	Decile calibration plots on hold-out sample showed good overall ordering. Minor deviations due to small group noise.

Appendix: Detailed Expense Considerations

Best Practices from Actuarial Literature:

- Expenses should not be modeled as a single percentage of premium; allocate by cost drivers (SOA 2012; CAS 2005)
- Acquisition/marketing → new business volume; underwriting/servicing → exposure; claims adjusting → claim counts/losses
- General overhead allocated separately to avoid distorting cost incidence (CAS DPP80-032)
- Consider policyholder persistence differences across risk classes (CAS DPP98-149)

Data Limitation & Modeling Decision:

Our dataset lacks operational expense drivers (new business volume, service workload, overhead allocation bases, policy persistence). Therefore, we apply a uniform expense loading consistently across all risk classes, reflecting expenses implicitly through the loading factor C = 1 / (1 - expense ratio - profit margin). Relative risk ordering is preserved through modeled pure premium.

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